

Aisha, 28 · 6 weeks since last period · positive home pregnancy test · sudden **right lower quadrant pain**, vaginal spotting · BP 100/65, HR 110 · beta-hCG 4,500 · transvaginal US: **no intrauterine pregnancy, complex mass in right adnexa** · **ectopic pregnancy in right uterine tube**

BIO 004 · WEEK 6 · CASE DEEP DIVE

Pregnancy and Birth

A positive pregnancy test, sharp lower-quadrant pain, and an embryo implanted where it can't survive.

PATIENT CASE

Aisha is 28, 6 weeks since her last menstrual period, with a **positive home pregnancy test** two weeks ago. This morning she developed **sharp right lower quadrant pain** and noticed **brown vaginal spotting**.

In the ED she is mildly tachycardic (HR 110). Beta-hCG is 4,500 mIU/mL — high enough that a normal intrauterine pregnancy should be visible on transvaginal ultrasound. Instead the scan shows **no gestational sac in the uterus** and a **complex mass in the right adnexa**. There is a small amount of free fluid in the cul-de-sac. The diagnosis is **ectopic pregnancy in the right uterine tube**; the team prepares for either medical management with methotrexate or surgical removal depending on stability and size.

Pregnancy is the female reproductive system's success story. Ectopic pregnancy is what happens when implantation occurs in the wrong place. To understand both, you have to walk from fertilization to implantation to placentation to birth.

GUIDING QUESTION

Where do fertilization and implantation happen, what does a placenta do, and what goes wrong when the embryo implants outside the uterus?

Where and how the sperm meets the ovum

Sperm deposited in the vagina swim through the cervix, traverse the uterus, and enter the uterine tubes. The ovum, released at ovulation and swept into the tube by fimbriae, meets sperm in the **ampulla of the uterine tube** — the wide section about midway along the tube. Capacitated sperm bind the zona pellucida; one penetrates; the zona reaction prevents others. Sperm and ovum nuclei fuse → **zygote**.

The zygote travels to the uterus

The zygote begins mitotic divisions while traveling slowly down the tube (3-4 days). It becomes a **morula** (solid ball), then a **blastocyst** (hollow ball with inner cell mass that becomes the embryo, and trophoblast that becomes placenta). The blastocyst reaches the uterus on day 5-6.

Implantation normally occurs in the upper uterine wall (anterior or posterior) around day 7. The trophoblast invades the endometrium and establishes the maternal-fetal interface.

Disposable organ shared by mother and fetus

The placenta develops from trophoblast + maternal decidua. By the end of the first trimester it has taken over hormonal and nutritional functions.

Chorionic villi

Finger-like projections of fetal tissue containing fetal capillaries. Bathed in maternal blood pooled in intervillous spaces.

Placental barrier

Fetal capillary endothelium + thin trophoblast layer separates fetal from maternal blood. Allows gas, nutrient, and waste exchange WITHOUT mixing blood.

Umbilical cord

Connects fetus to placenta. Contains *two umbilical arteries* (carry deoxygenated blood from fetus to placenta) and *one umbilical vein* (oxygenated blood to fetus). Wharton's jelly cushions the vessels.

Hormones

hCG (maintains corpus luteum early), estrogen, progesterone (maintain pregnancy), hPL (alters maternal metabolism for fetal benefit).

Three stages of fetal development

TRIMESTER	WEEKS	KEY EVENTS
First	1-12	Organogenesis. Most vulnerable to teratogens. Heartbeat by week 6 on transvaginal US.
Second	13-27	Rapid growth, maternal "quickening" (feeling fetal movement) ~16-20 weeks. Anatomic survey US around 20 weeks.
Third	28-40	Growth and maturation. Surfactant maturation in lungs (~34 weeks). Viability improves week by week.

Three stages of birth

● Stage 1 — Dilation

Longest. Cervix dilates 0 → 10 cm. Latent (slow) and active (faster) phases. Regular uterine contractions.

● Stage 2 — Expulsion

Full dilation to delivery of baby. Maternal pushing + uterine contractions. Cardinal movements: engagement, descent, flexion, internal rotation, extension (head out), external rotation (restitution), expulsion of body.

● Stage 3 — Placenta

Delivery of placenta. Usually within 30 minutes of baby. Uterus contracts to control bleeding from where placenta attached.

Labor is driven by progressive increase in oxytocin, fetal cortisol, and prostaglandins. The cervix softens and effaces under prostaglandin influence; uterine contractions intensify with oxytocin.

What changes after delivery

The uterus involutes from ~1 kg to pre-pregnancy size over 6 weeks. Lactation begins as estrogen and progesterone drop and prolactin rises (the placenta's estrogen had been blocking prolactin's effect on the breast). Milk production is established within days; oxytocin (released in response to nipple stimulation) drives the let-down reflex.

Maternal cardiovascular changes (increased blood volume, increased cardiac output, decreased SVR) gradually reverse over weeks.

DRAW ZONE 1 -- Diagram fertilization through implantation: sperm in vagina, fertilization in ampulla, cleavage, morula, blastocyst, normal uterine implantation. Then mark the abnormal sites where ectopic pregnancy can occur.

98% of ectopics are tubal (most often ampulla). Other sites: ovary, abdomen, cervix.



DRAW ZONE 2 -- Sketch the placenta and fetal-maternal interface. Show chorionic villi, intervillous space, umbilical vessels (2 arteries, 1 vein), and the placental barrier.

Color fetal vessels and maternal blood differently. Star the barrier — it separates without mixing.

An embryo in the wrong place

Ectopic pregnancy is the leading cause of first-trimester maternal death. Aisha's tubal pregnancy can rupture as it grows, with catastrophic hemorrhage. Identifying it early saves her life and often her fertility.

"RLQ pain + bleeding"

Implantation in the tube stretches and inflames the tubal wall, causing pain. Hormone changes and local irritation cause spotting/bleeding from the uterine endometrium.

"Beta-hCG 4,500 with no IUP visible"

Transvaginal US should detect a gestational sac at beta-hCG > 1,500-2,000 mIU/mL. Absence at 4,500 + adnexal mass = ectopic until proven otherwise.

"Why most ectopics are tubal"

Tubal damage (PID, prior ectopic, prior surgery, smoking) impairs cilia and tube motility. The embryo can't complete its journey to the uterus and implants where it stops.

"Risk of rupture"

The tube has limited capacity to expand. As the embryo grows it stretches the tube. Rupture causes massive intra-abdominal bleeding and shock. Aisha's mild tachycardia and free fluid suggest she may be early in this process — surgical urgency.

"Methotrexate vs surgery"

Methotrexate (folate antagonist; halts trophoblast division) is an option in stable patients with small (<3.5 cm) ectopics and beta-hCG <5,000. Surgical removal (salpingostomy or salpingectomy) for larger ectopics, ruptured ectopics, or unstable patients.

"Future fertility"

Most women conceive successfully after an ectopic, especially if the contralateral tube is healthy. Risk of recurrent ectopic is increased (~15%).

Tube, implantation, placenta, blood supply. Aisha's case is the reproductive system's most dangerous misstep — a fertilized ovum implanting where the body cannot support it.

